CHANDU PRASAD BHAIRAPU



AI Engineer

bhairapuchanduprasad@gmail.com (+49) 15560194877 Berlin • EU Blue Card linkedin github

ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in LLM and embedding steps inside data pipelines, from structured extraction and retrieval to evaluation against ground-truth data. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by a strong foundation in Python (pandas), SQL, data analysis and machine learning across enterprise and start-up environments.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer owning the full knowledge-extraction data pipeline end-to-end: data ingestion, chunking, prompt engineering and indexing (agentic, deterministic and monolithic strategies), reporting directly to the CTO and working with SAP domain experts across the German energy sector.
- Designed and optimized multi-agent structured extraction with LangGraph (entity-extraction, relationship-analysis, consolidation and reflection agents running in parallel) over complex regulatory and technical documents, raising precision/recall from 33%/45% to 93%/90% in evaluation against an SAP-system-derived ground truth.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Designed and implemented a Model Context Protocol (MCP) query service exposing the knowledge graph for millisecond-latency retrieval, integrated directly into Claude Code for agentic development workflows.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) deployed on Azure, where domain experts validate and approve knowledge-graph changes before they are written.
- Delivered as an AI-augmented engineer, using Claude Code, custom skills and agentic coding workflows under clean code principles.
- **Tech stack: Python, Postgres, LangGraph, LangChain, LiteLLM, RAG, GraphRAG, Vector Search, MCP, MongoDB, Neo4j, FastAPI, React, Docker, Kubernetes, Azure**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Designed and deployed an agentic AI application with LangGraph that routes each request across vector database queries, general LLM inference and summarization, after mapping business stakeholders' decision-making workflows.
- Developed a Python backend with FastAPI and a React front-end, using Postman to check API behaviour.
- Built and containerized the entire stack with Docker and deployed it to AWS using CDK, following clean code practices; version-controlled in GitLab with CI/CD.
- **Tech stack: Python, FastAPI, REST APIs, LangGraph, LangChain, React, Postman, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated RAG generation and retrieval evaluation with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.
- Designed RAG pipelines, focusing on chunking strategies, vector databases, indexing, and fine-tuning models for optimized retrieval.
- Built AWS cloud infrastructure with Lambda, SQS and DynamoDB for PrismaChat RAG evaluation using CDK.

- **Tech stack: Python, Embeddings, Hugging Face Transformers, LangChain, PyTorch, RAGAS, PytrecEval, Vector Databases, AWS (Lambda, SQS, DynamoDB, CDK), Docker**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity, reaction progression) from time-series process data, evaluating tree-based vs. deep learning approaches for this tabular/temporal hybrid problem.
- Engineered lag features and rolling statistical windows from raw sensor time-series (temperature, reaction time, process variables) with pandas to feed the tree-based models, while feeding raw sequences directly into the LSTM.
- Benchmarked model performance against traditional empirical curve-fitting methods, improving early identification of batches likely to deviate from target specification.
- Deployed an interactive Streamlit dashboard exposing model predictions to chemists in real time, letting them simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, pandas, NumPy, Scikit-Learn, XGBoost, TensorFlow/Keras, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XG-Boost regressor, FB Prophet, LSTM models for time series analysis for EVs.
- Developed LSTM, achieved 96% test accuracy and strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: Spin-off Ecosystem Data Hub | Python | Postgres | Embeddings | LLMs

- Central Postgres data model joining a mock CRM (two-way sync over REST APIs and webhooks, no duplicates, alerts on failed syncs) with publication and patent data pulled from public APIs and scraping, keeping a source and timestamp for every field so each result shows where it comes from.
- Entity resolution of people and organisations (rules, then embeddings, then LLM structured extraction for unclear cases) with a labelled reference set, precision/recall evaluation in GitHub Actions, and GDPR (DSGVO) consent and deletion handling built into the data model.

TECHNICAL SKILLSET

- **Data:** Python, SQL, Postgres, pandas, NumPy, notebooks, MongoDB, Neo4j, Vector Databases (Qdrant, Pinecone, Azure AI Search)
- **LLMs & AI:** LLMs, Embeddings, structured extraction, LangGraph, LangChain, LiteLLM, RAG, GraphRAG, MCP, evaluation (RAGAS, PytrecEval), Hugging Face Transformers, QLoRA fine-tuning
- **Machine Learning:** Scikit-Learn, XGBoost, PyTorch, TensorFlow, classification and regression, time series
- **APIs & Cloud:** FastAPI, REST APIs, Postman, Docker, Kubernetes, AWS (Lambda, SQS, DynamoDB, S3, EC2, CDK), Azure, Git, GitLab CI/CD, Claude Code, Agile
- **Languages:** English (Business Fluent), German (B1), Hindi (Fluent), Telugu (Native)

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering (Minor in Computer Science & Engineering) — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI